LECTURE 20 (153-248)

MA(q) $\{y_t\}$

MA(0) → y_t : $y_t \overset{iid}{\sim} N(\mu, \sigma^2)$

$y_t = \mu + \varepsilon_t$ where $\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

Slutzky

$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$

→ MA(q)

$\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

$\mu, \theta_1 \cdots \theta_q, \sigma$

$\mathbb{E} y_t = \mu$

Applied to preprocessed data (differencing or AR residuals).

ARIMA(p, d, q) →

p: 0

d: order of differencing

→ KALMAN FILTER

MA(1)

$y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

Likelihood

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix}, \underset{n \times n}{\Sigma} \right)$$

$$\Sigma = \begin{bmatrix} & & 0 \\ & \ddots & \\ 0 & & \end{bmatrix}_{n \times n}$$

$\text{var}(y_t) = \sigma^2(1+\theta^2)$

$\text{Cov}(y_t, y_{t+1}) = \theta\sigma^2$

$$\frac{1}{\sqrt{\det \Sigma}} \exp\left[- \frac{(y-\mu)^T \Sigma^{-1} (y-\mu)}{2} \right]$$

Want to write likelihood more explicitly.

Conditional Trick to write an approximate likelihood.

ε_t iid $N(0, \sigma^2)$

$$y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad t = 1, \ldots, n.$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad \begin{pmatrix} \varepsilon_0 \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

Set $\varepsilon_0 = 0$ → TRICK ASSUMPTION

$$f_{y_1 \ldots y_n}(y_1, \ldots y_n)$$

$$= f_{\varepsilon_1 \ldots \varepsilon_n}\big(y_1 - \mu,\ y_2 - \mu - \theta(y_1 - \mu),\ y_3 - \mu - \theta(y_2 - \mu) + \theta^2(y_1 - \mu),\ \ldots\big)$$

$y_1 = \mu + \varepsilon_1 + \theta\varepsilon_0$

$= \mu + \varepsilon_1$

$\Rightarrow \varepsilon_1 = y_1 - \mu$

$y_2 = \mu + \varepsilon_2 + \theta\varepsilon_1$

$= \mu + \varepsilon_2 + \theta(y_1 - \mu)$

$\Rightarrow \varepsilon_2 = (y_2 - \mu) - \theta(y_1 - \mu)$

$$\varepsilon_3 = y_3 - \mu - \theta\varepsilon_2$$
$$= y_3 - \mu - \theta[y_2 - \mu] + \theta^2(y_1 - \mu)$$

$$\varepsilon_n = y_n - \mu - \theta(y_{n-1} - \mu) + \theta^2(y_{n-2} - \mu) - \cdots (y_1 - \mu)$$

$$\hat{\varepsilon}_1 = y_1 - \mu$$
$$\hat{\varepsilon}_2 = y_2 - \mu - \theta(y_1 - \mu) = y_2 - \mu - \theta\hat{\varepsilon}_1$$
$$\hat{\varepsilon}_t = y_t - \mu - \theta\hat{\varepsilon}_{t-1}, \quad t = 2, \ldots, n$$

Likelihood: $f_{y_1 \cdots y_n}(y_1 \cdots y_n)$

$$= f_{\varepsilon_1 \cdots \varepsilon_n}(\hat{\varepsilon}_1 \cdots \hat{\varepsilon}_n)$$

$$= \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{\hat{\varepsilon}_j^2}{2\sigma^2}\right]$$

$$\propto \sigma^{-n} \exp\left[-\frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^2}{2\sigma^2}\right]$$

$$= \sigma^{-n} \exp\left[-\frac{S(\mu, \theta)}{2\sigma^2}\right]$$

$$S(\mu, \theta) = \sum \hat{\varepsilon}_i^2$$
$$= [y_1 - \mu]^2 + [(y_2 - \mu) - \theta(y_1 - \mu)]^2$$

$$+ \left[(y_3 - \mu) - \theta(y_2 - \mu) + \theta^2(y_1 - \mu)\right]$$

$$+ \cdots + \left[(y_n - \mu) - \theta(y_{n-1} - \mu) + \theta^2(y_{n-2} - \mu)^2 - \cdots\right]$$

$(\hat{\mu}, \hat{\theta})$ = minimizer of $S(\mu, \theta)$

(initialize at $\mu_{init} = \bar{y}$, $\theta_{init} = 0$)

$$\hat{\sigma} = \sqrt{\frac{S(\hat{\mu}, \hat{\theta})}{n}}$$

Standard Errors for $\hat{\mu}, \hat{\theta}$

Likelihood: $\sigma^{-n} \exp\left(-\frac{S(\mu, \theta)}{2\sigma^2}\right)$

prior: $\mu, \theta, \log\sigma \sim \text{Unif}(-C, C)$

$\Rightarrow f(\mu, \theta, \sigma) \propto \frac{1}{\sigma} I(\sigma > 0)$

μ, θ, σ

posterior: $\propto \sigma^{-n-1} \exp\left(-\frac{S(\mu, \theta)}{2\sigma^2}\right) I(\sigma > 0)$

Integrate w.r.t σ:

Posterior of μ, θ: $\left[\frac{S(\hat{\mu}, \hat{\theta})}{S(\mu, \theta)}\right]^{n/2} \rightarrow \left(\frac{1}{S(\mu, \theta)}\right)^{n/2}$

Previous: $\left(\frac{1}{S(\beta)}\right)^{n/2}$

not a square in μ, θ

$\alpha = (\mu, \theta)$

$$\approx \left[\frac{S(\hat{\mu}, \hat{\theta})}{S(\hat{\mu}, \hat{\theta}) + \frac{1}{2} (\alpha - \hat{\alpha})^T HS(\hat{\alpha}) (\alpha - \hat{\alpha})} \right]^{n/2}$$

$$HS(\hat{\alpha}) = HS(\hat{\mu}, \hat{\theta}) = \begin{pmatrix} \frac{\partial^2 S}{\partial \mu^2} & \frac{\partial^2 S}{\partial \mu \partial \theta} \\ \frac{\partial^2 S}{\partial \mu \partial \theta} & \frac{\partial^2 S}{\partial \theta^2} \end{pmatrix}$$

$$\alpha \mid \text{data} \sim t\left(\hat{\alpha}, \frac{S(\hat{\alpha})}{n-2} \left(\frac{1}{2} HS(\hat{\alpha}) \right)^{-1} \right)$$

t: dimension 2, df: $n-2$

$$\left[\frac{1}{1 + \frac{1}{\kappa} (x-m)^T \Sigma^{-1} (x-m)} \right]^{\frac{\kappa+p}{2}}$$

Standard errors: $S(\mu, \theta)$

Full likelihood for MA(1)

ARIMA

KALMAN FILTER

$y_1 \dots y_n$

$$y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1} \qquad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$$

$$f_{y_1 \cdots y_n \mid \mu, \theta, \sigma}(y_1 \cdots y_n)$$

$$= f_{y_1}(y_1)\, f_{y_2 \mid y_1}(y_2)\, f_{y_3 \mid y_1, y_2}(y_3) \cdots f_{y_t \mid y_1 \cdots y_{t-1}}(y_t) \cdots f_{y_n \mid y_1 \cdots y_{n-1}}(y_n)$$

$$f_{y_t \mid y_1 \cdots y_{t-1}}(y_t) \longrightarrow \text{density of } N(a_t, V_t)$$

$$a_t = \mathbb{E}(y_t \mid y_1 \cdots y_{t-1})$$

$$V_t = \operatorname{var}(y_t \mid y_1 \cdots y_{t-1})$$

a_1, V_1

$$a_1 = \mathbb{E}(y_1) = \mathbb{E}(\mu + \varepsilon_1 + \theta\varepsilon_0) = \mu$$

$$V_1 = \operatorname{var}(y_1) = \operatorname{var}(\mu + \varepsilon_1 + \theta\varepsilon_0) = \sigma^2 + \theta^2\sigma^2 = \sigma^2(1+\theta^2)$$

$$\mathbb{E}(y_t \mid y_1 \cdots y_{t-1})$$

Trick: Figure out:

$$\begin{pmatrix} y_t \\ y_{t-1} \end{pmatrix} \Big| \, y_1 \cdots y_{t-2} \sim N\left(\begin{matrix} \mu \\ \boxed{\mathbb{E}(y_t \mid y_1 \cdots y_{t-2})} \\ \mathbb{E}(y_{t-1} \mid y_1 \cdots y_{t-2}) \end{matrix} \right.$$

a_{t-1}

$$E(y_t \mid y_1 \cdots y_{t-2})$$

$$= E(y_t) = \mu$$

$$\begin{pmatrix} y_t \\ y_{t-1} \end{pmatrix} \Big| y_1 \cdots y_{t-2} \sim N\left(\begin{pmatrix} \mu \\ a_{t-1} \end{pmatrix}, \begin{pmatrix} \sigma^2(1+\theta^2) & \theta\sigma^2 \\ \theta\sigma^2 & V_{t-1} \end{pmatrix} \right)$$

$$\operatorname{var}(y_t \mid y_1 \cdots y_{t-2})$$

$$= \operatorname{var}(y_t) = \sigma^2(1+\theta^2)$$

$$\operatorname{var}(y_{t-1} \mid y_1 \cdots y_{t-2}) = V_{t-1}$$

$$\operatorname{Cov}(y_t, y_{t-1} \mid y_1 \cdots y_{t-2})$$

$$= \operatorname{Cov}(\mu + \varepsilon_t + \theta\varepsilon_{t-1}, \mu + \varepsilon_{t-1} + \theta\varepsilon_{t-2} \mid y_1 \cdots y_{t-2})$$

$$= \operatorname{Cov}(\varepsilon_t + \theta\varepsilon_{t-1}, \varepsilon_{t-1} + \theta\varepsilon_{t-2} \mid y_1 \cdots y_{t-2})$$

$$= \operatorname{Cov}(\theta\varepsilon_{t-1}, \varepsilon_{t-1} \mid y_1 \cdots y_{t-2})$$

$$= \operatorname{Cov}(\theta\varepsilon_{t-1}, \varepsilon_{t-1}) = \theta\sigma^2$$

We have proved:

$$\begin{pmatrix} y_t \\ y_{t-1} \end{pmatrix} \Big| y_1 \cdots y_{t-2} \sim N\left(\begin{pmatrix} \mu \\ a_{t-1} \end{pmatrix}, \begin{pmatrix} \sigma^2(1+\theta^2) & \theta\sigma^2 \\ \theta\sigma^2 & V_{t-1} \end{pmatrix} \right)$$

$$y_t \mid y_1, \ldots y_{t-1}$$

$$\sim N\left(\mu + \frac{\theta\sigma^2}{V_{t-1}}(y_{t-1} - a_{t-1}),\ \sigma^2(1+\theta^2) - \frac{(\theta\sigma^2)^2}{V_{t-1}}\right)$$

Recursion:

$$a_t = \mu + \frac{\theta\sigma^2}{V_{t-1}}(y_{t-1} - a_{t-1})$$

$$V_t = \sigma^2(1+\theta^2) - \frac{(\theta\sigma^2)^2}{V_{t-1}}$$

$t = 2, 3, \ldots$

$$a_1 = \mu, \quad V_1 = \sigma^2(1+\theta^2)$$

Likelihood:

$$\prod_{t=1}^{n} \frac{1}{\sqrt{2\pi}\sqrt{V_t}} \exp\left(-\frac{(y_t - a_t)^2}{2V_t}\right)$$